

# VMAX<sup>3</sup> Hardware Fundamentals



Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

1

## Welcome to the VMAX<sup>3</sup> Hardware Fundamentals.

Copyright © 1996, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014 EMC Corporation. All Rights Reserved. EMC believes the information in this publication is accurate as of its publication date. The information is subject to change without notice.

THE INFORMATION IN THIS PUBLICATION IS PROVIDED "AS IS." EMC CORPORATION MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WITH RESPECT TO THE INFORMATION IN THIS PUBLICATION, AND SPECIFICALLY DISCLAIMS IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Use, copying, and distribution of any EMC software described in this publication requires an applicable software license. EMC2, EMC, Data Domain, RSA, EMC Centera, EMC ControlCenter, EMC LifeLine, EMC OnCourse, EMC Proven, EMC Snap, EMC SourceOne, EMC Storage Administrator, Acartus, Access Logix, AdvantEdge, AlphaStor, ApplicationXtender, ArchiveXtender, Atmos, Authentica, Authentic Problems, Automated Resource Manager, AutoStart, AutoSwap, AVALONidm, Avamar, Captiva, Catalog Solution, C-Clip, Celerra, Celerra Replicator, Centera, CenterStage, CentraStar, ClaimPack, ClaimsEditor, CLARiiON, ClientPak, Codebook Correlation Technology, Common Information Model, Configuration Intelligence, Configuresoft, Connectrix, CopyCross, CopyPoint, Dantz, DatabaseXtender, Direct Matrix Architecture, DiskXtender, DiskXtender 2000, Document Sciences, Documentum, eInput, E-Lab, EmailXaminer, EmailXtender, Enginuity, eRoom, Event Explorer, FarPoint, FirstPass, FLARE, FormWare, Geosynchrony, Global File Virtualization, Graphic Visualization, Greenplum, HighRoad, HomeBase, InfoMover, Infospace, Infra, InputAccel, InputAccel Express, Invista, Ionix, ISIS, Max Retriever, MediaStor, MirrorView, Navisphere, NetWorker, nLayers, OnAlert, OpenScale, PixTools, Powerlink, PowerPath, PowerSnap, QuickScan, Rainfinity, RepliCare, RepliStor, ResourcePak, Retrospect, RSA, the RSA logo, SafeLine, SAN Advisor, SAN Copy, SAN Manager, Smarts, SnapImage, SnapSure, SnapView, SRDF, StorageScope, SupportMate, SymmAPI, SymmEnabler, Symmetrix, Symmetrix DMX, Symmetrix VMAX, TimeFinder, UltraFlex, UltraPoint, UltraScale, Unisphere, VMAX, Vblock, Viewlets, Virtual Matrix, Virtual Matrix Architecture, Virtual Provisioning, VisualSAN, VisualSRM, Voyence, VPLEX, VSAM-Assist, WebXtender, xPression, xPresso, YottaYotta, the EMC logo, and where information lives, are registered trademarks or trademarks of EMC Corporation in the United States and other countries.

All other trademarks used herein are the property of their respective owners.

© Copyright 2014 EMC Corporation. All rights reserved. Published in the USA.

Revision Date: October 2014

Revision Number: MR-7WP-VMAXHDFD

# Course Overview

|                    |                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | This course is a hardware fundamental training focused on the VMAX <sup>3</sup> Family including VMAX 100K, 200K and 400K models. The intention of this course is to be used as prerequisite training for those whom are going to attend the VMAX <sup>3</sup> Installation and Maintenance Course and those seeking to acquire the Platform Engineer Specialist Certification |
| <b>Audience</b>    | This course is intended for those responsible for the design, implementation and operation of the VMAX <sup>3</sup> Family of arrays and those seeking to attend the VMAX <sup>3</sup> Installation and Maintenance Course .                                                                                                                                                   |
| <b>Objectives</b>  | <p>Upon completion of this course, you should be able to:</p> <ul style="list-style-type: none"><li>• List the VMAX<sup>3</sup> models</li><li>• Explain the VMAX<sup>3</sup> hardware architecture</li><li>• Describe the VMAX<sup>3</sup> System Bay Components</li><li>• Access the system</li></ul>                                                                        |

This course covers the VMAX<sup>3</sup> models, and explanation of the VMAX<sup>3</sup> hardware architecture, a description of the VMAX<sup>3</sup> System Bay components and ways to access the system.

## Module 1: VMAX<sup>3</sup> Hardware Fundamentals Introduction

Upon completion of this module, you should be able to:

- Describe the VMAX<sup>3</sup> with HYPERMAX OS 5977 array offering
- Compare models
- List common features



This module focuses on an introduction to the three VMAX<sup>3</sup> models: the VMAX 100K, 200K and 400K. We will also compare and contrast these models.

# Model Identification

| Prod ID/<br>S/N Identifier | GA Model Name |
|----------------------------|---------------|
| x968                       | VMAX 100K     |
| x967                       | VMAX 200K     |
| x972                       | VMAX 400K     |

Factory Pre-configured based on SLOs



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

4

VMAX<sup>3</sup> with HYPERMAX OS (previously known as Enginuity) 5977 arrays include the VMAX 100K for Enterprise and commercial data centers, the VMAX 200K for most Enterprise data centers, and the VMAX 400K for large-environment Enterprise data centers. For high-demand storage environments, where extremely low latency and high IOPS are required, the VMAX<sup>3</sup> Family can also be configured with all Flash. VMAX<sup>3</sup> arrays are pre-configured with array-based software and hardware configurations based on pre-packaged Service Level Objectives (SLOs).

# VMAX 100K

## SYSTEM FEATURES

- Supports 1-2 Engines
- Up to 1440 2.5" Drives
- Up to 720 3.5" Drives
- Up to 0.5 PBU Capacity
- Up to 64 FE Ports
- Infiniband Fabric
  - ▶ Dual 12-Port Switches



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

5

The VMAX 100K is configured with one to two engines. With the maximum two-engine configuration, the VMAX 100K supports up to (1440) 2.5" drives, or up to (720) 3.5" drives, providing up to 0.5 Petabytes of usable capacity. When fully configured, the 100K provides up to 64 front-end ports for host connectivity. The internal fabric interconnect uses dual Infiniband 12-port switches for redundancy and availability.

# VMAX 200K

## SYSTEM FEATURES

- Supports 1-4 Engines
- Up to 2880 2.5" Drives
- Up to 1440 3.5" Drives
- Up to 2.1 PBU Capacity
- Up to 128 FE Ports
- Infiniband Fabric
  - ▶ Dual 12-Port Switches



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

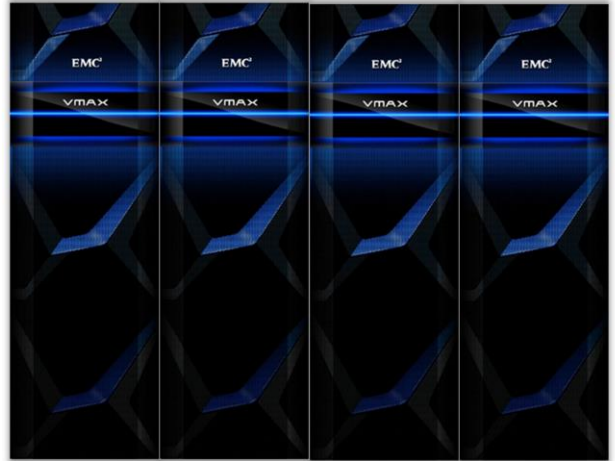
6

The VMAX 200K is configured with one-to-four engines. With the maximum four-engine configuration, the VMAX 200K supports up to (2880) 2.5" drives, or up to (1440) 3.5" drives, providing up to 2.1 Petabytes of usable capacity. When fully configured, the 200K provides up to 128 front-end ports for host connectivity. The internal fabric interconnect uses dual Infiniband 12-port switches for redundancy and availability.

# VMAX 400K

## SYSTEM FEATURES

- Supports 1-8 Engines
- Up to 5760 2.5" Drives
- Up to 2880 3.5" Drives
- Up to 4 PBU Capacity
- Up to 256 FE Ports
- Infiniband Fabric
  - ▶ Dual 18-Port Switches



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

7

The VMAX 400K is configured with one to eight engines. With the maximum eight-engine configuration, the VMAX 400K supports up to (5760) 2.5" drives, or up to (2880) 3.5" drives, providing up to 4 Petabytes of usable capacity. When fully configured, the 400K provides up to 256 front-end ports for host connectivity. The internal fabric interconnect uses dual Infiniband 18-port switches for redundancy and availability.

## Model Comparison

|                     | VMAX 100K             | VMAX 200K             | VMAX 400K             |
|---------------------|-----------------------|-----------------------|-----------------------|
| # of Engines        | 1-2                   | 1-4                   | 1-8                   |
| Cache/Engine        | 512GB or 1TB          | 512GB, 1TB, or 2TB    | 512GB, 1TB, or 2TB    |
| Max 2.5" Drives     | 1440                  | 2880                  | 5760                  |
| Max 3.5" Drives     | 720                   | 1440                  | 2880                  |
| Max Usable Capacity | 0.5 P Bu              | 2.1 P Bu              | 4.0 P Bu              |
| Max FE Ports        | 64                    | 128                   | 256                   |
| Fabric              | Dual 12-Port switches | Dual 12-Port switches | Dual 18-Port switches |



Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

8

This table shows a comparison of all three VMAX<sup>3</sup> Family arrays. Please take a moment to review them.

## Common Features

| VMAX 100K, 200K, 400K     |                                                                                                                                                                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feature                   | Description                                                                                                                                                                                                                                                                  |
| Maximum Drives per Engine | <ul style="list-style-type: none"> <li>• (720) 2.5"</li> <li>• (360) 3.5"</li> </ul>                                                                                                                                                                                         |
| Drive Options             | <ul style="list-style-type: none"> <li>• Hybrid (mixed drive types)</li> <li>• All-Flash</li> </ul>                                                                                                                                                                          |
| DAE Mixing                | <ul style="list-style-type: none"> <li>• 60-drive and 120-drive behind an engine</li> <li>• Single increments</li> </ul>                                                                                                                                                     |
| Power Options             | <ul style="list-style-type: none"> <li>• Three-Phase Delta (50 amp), WYE (32 amp)</li> <li>• Single-Phase (32 amp)</li> </ul>                                                                                                                                                |
| Dispersion                | <ul style="list-style-type: none"> <li>• System Bays 2-8, up to 25 meters from System Bay 1</li> </ul>                                                                                                                                                                       |
| Vault                     | <ul style="list-style-type: none"> <li>• Vault to Flash in engine</li> </ul>                                                                                                                                                                                                 |
| Racking Options           | <ul style="list-style-type: none"> <li>• Single engine</li> <li>• Dual engine</li> <li>• Third-party racking</li> </ul>                                                                                                                                                      |
| Service Access            | <ul style="list-style-type: none"> <li>• Integrated service processor in System Bay 1                             <ul style="list-style-type: none"> <li>• Management Module Control Station (MMCS)</li> </ul> </li> <li>• Service tray on additional system bays</li> </ul> |

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

9

Common features throughout the VMAX<sup>3</sup> Family include maximum drives per engine; both hybrid and all-Flash, DAE mixing behind engines in single increments, power configuration options, system bay dispersion, multiple racking options and service access points. Also, Vault to Flash in engine is implemented on the with VMAX<sup>3</sup> Family, which is a change from the previous vaulting process. Service access is provided by a Management Module Control Station (MMCS), which is the integrated service processor located in System Bay 1.

This table shows the common features between the VMAX 100K, 200K and 400K. Please take a moment to review them.

## Module 1: Summary

Key points covered in this module:

- Describe the VMAX<sup>3</sup> with HYPERMAX OS 5977 array offering
- Compare models
- List common features



This module covered an introduction to the three VMAX<sup>3</sup> models: the VMAX 100K, 200K and 400K and a model comparison.

## Module 2: Hardware Components

Upon completion of this module, you should be able to:

- Identify hardware components

This module focuses on VMAX<sup>3</sup> with HYPERMAX OS 5977 hardware components locations and functionality.

# Module 2: Hardware Components

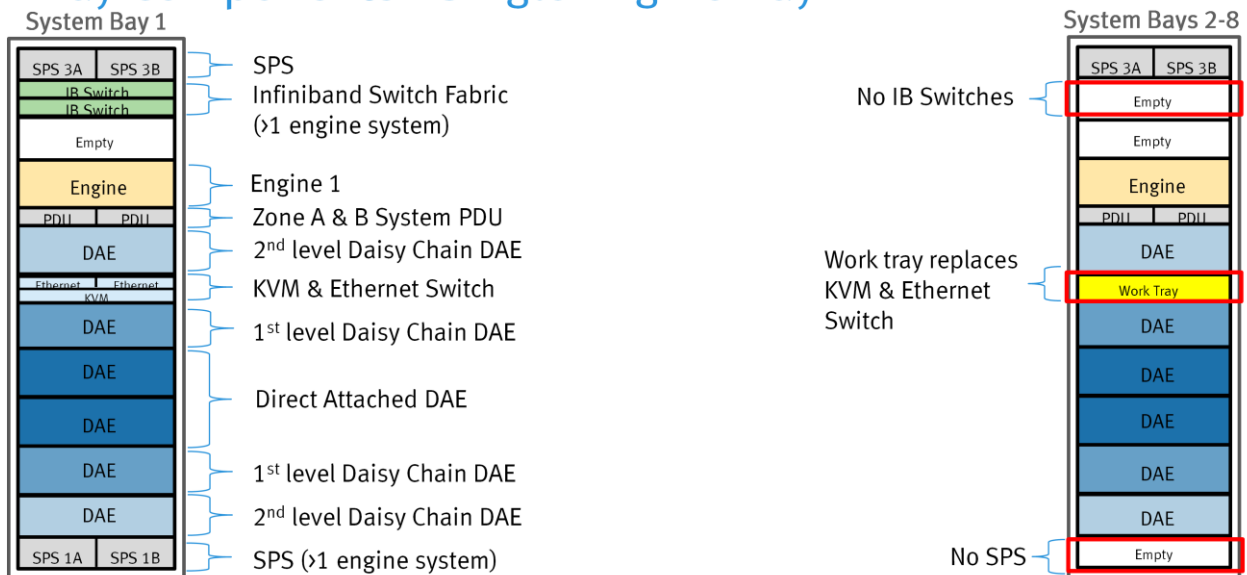
## Lesson 1: System Bays

This lesson covers the following topics:

- Single Engine Bay Components
- Dual Engine Bay Components

This lesson covers System Bay component locations.

## Bay Components - Single Engine Bay



EMC<sup>2</sup>

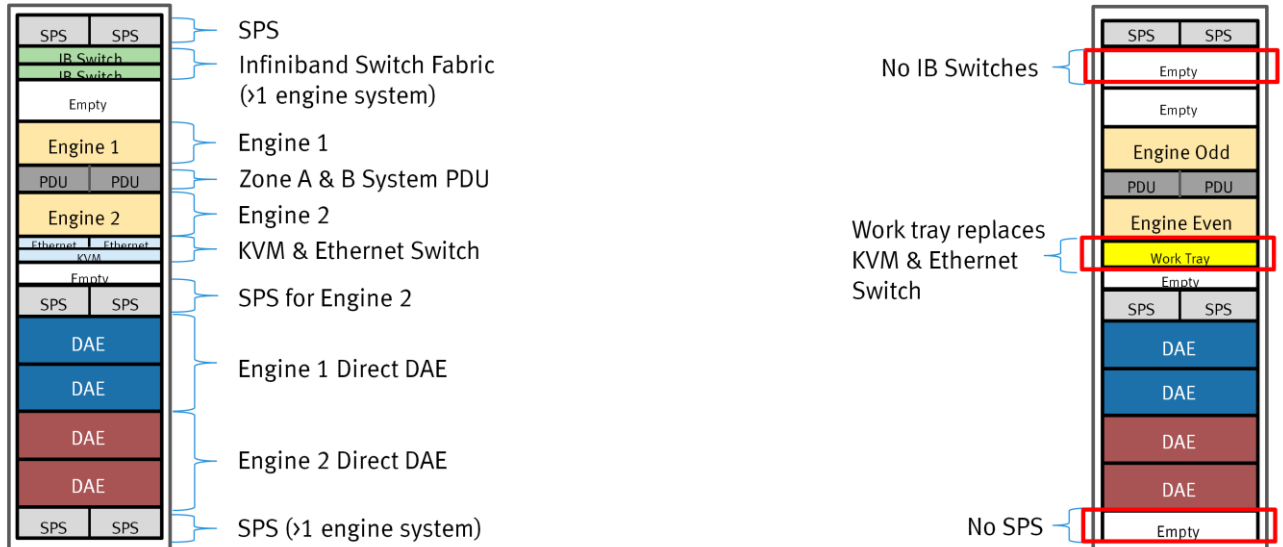
Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

13

Components in System Bay 1 of a single engine bay include Standby Power Supplies (SPSs) to support the engine, a Fabric and additional SPS (in multi-engine systems only), one engine, a Power Distribution Unit (PDU), a KVM (Keyboard, Video, Mouse), a set of Ethernet switches (seen from the rear of the bay) and DAEs associated with the engine. System Bays 2 through 8 do not have the Fabric and additional SPSs, KVM or Ethernet switches. A Work Tray is included in System Bays 2 through 8 in place of the KVM. We will discuss these components in more detail in upcoming slides.

## Bay Components - Dual Engine Bay



Components in System Bay 1 of a dual engine bay include Standby Power Supplies (SPSs) to support the engines, a Fabric and SPSs to support it (in multi-engine systems only), one or two engines, a Power Distribution Unit (PDU), a KVM (Keyboard, Video, Mouse), an Ethernet switch and DAEs. System Bays 2 through 4 do not have the Fabric and its SPSs, a KVM or a set of Ethernet switches. A Work Tray is included in System Bays 2 through 4 in place of the KVM. We will discuss these components in more detail in upcoming slides.

## Li Ion Standby Power Supply (SPS) - Single Engine Bays



\* Present in multi-engine systems only

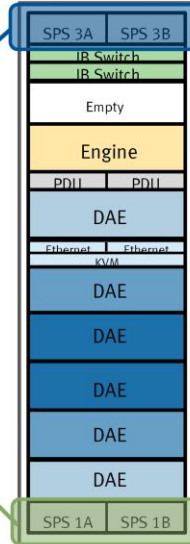
### SPS 3A/3B

Power and Battery Backup for Engine (and Fabric in multi-engine systems)

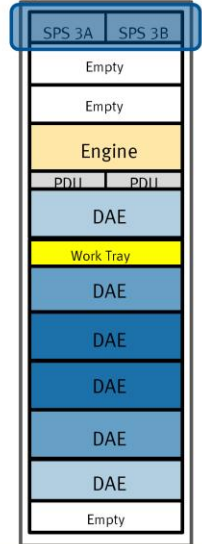
### SPS 1A/1B\*

Power and Battery Backup for Fabric & Engine (System Bay 1 only)

### System Bay 1



### System Bays 2-8



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

15

The Li Ion SPS is a FRUable 10-lb battery, removed from the front of the system. The SPS transports separately and will be part of the install procedure. SPS units have 2.2KW capacity, and provide two five-minute intervals of battery life to allow vaulting in the event of a power failure. In System Bay 1 of a single engine bay, there are up to two SPS trays (four SPSs total) to support the engine and Fabric, if installed. In System Bays 2 through 8, there is only one SPS tray (two SPSs) to support the engine.

When installed in System Bay 1, SPS 1A and 1B are cabled to both the engine and the fabric, and SPS 3A and 3B are also cabled to both the engine and the fabric. This is true in multi-engine systems only. In single engine systems, SPS 3A and 3B support the engine only.

## Li Ion Standby Power Supply (SPS) - Dual Engine Bays



\*Present in multi-engine systems only  
 \*\*Supports second engine (if installed)

### SPS 3A/3B

Power and Battery Backup for Odd Engine (and Fabric in System Bay 1 in multi-engine systems)

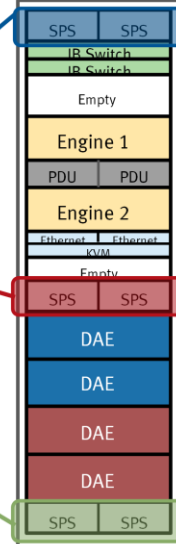
### SPS 2A/2B\*\*

Power and Battery Backup for Even Engine

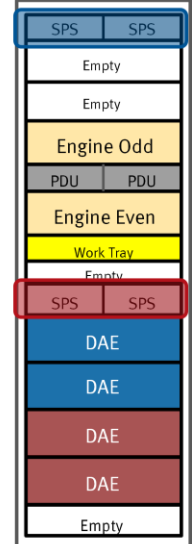
### SPS 1A/1B\*

Power and Battery Backup for Fabric & Engine (System Bay 1 only)

### System Bay 1



### System Bays 2-4



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

16

In System Bay 1 of a dual engine bay, there are up to three SPS trays (six SPSs total) to support the engines and Fabric, if installed. In System Bays 2 through 4, there are up to two SPS trays (four SPSs total) to support the engines.

When installed in System Bay 1, SPS 1A and 1B are cabled to both the engine and the fabric, and SPS 3A and 3B are also cabled to both the engine and the fabric. This is true in multi-engine systems only. In single engine systems, SPS 3A and 3B support the engine only.

## Bay Components - Fabric\*

### Infiniband Switch (Fabric)\*

System Bay 1 only (two switches)  
56 Gbps/port  
Carries control, metadata, user data  
Two Types

- 12-Port (VMAX 100K and 200K)
- 18-Port (VMAX 400K)

Connects to every director



12-Port Rear



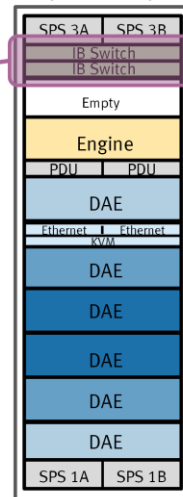
18-Port Front



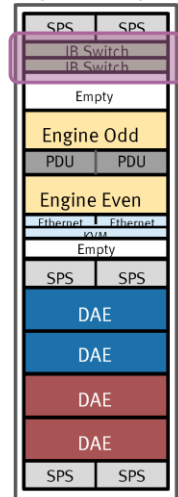
18-Port Rear

\*Present in multi-engine systems only

### Single Engine System Bay 1



### Dual Engine System Bay 1



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

17

Fabric interconnect is provided by dual Infiniband switches, also known as the Fabric. The Fabric is present in multi-engine systems only, and installed in System Bay 1. Standby Power Supplies (SPSs) are used for battery backup to the Fabric in the event of a power failure.

There are two types of Fabrics; a pair of 18-port switches used in the VMAX 400K, and a pair of 12-port switches used in the VMAX 100K and VMAX 200K. The 18-port switches have redundant, hot-pluggable power supplies and a fan assembly, all of which are FRUs. The 12-port switches have dual power supplies, which are not FRUs. In the event of a power supply failure on a 12-port switch, the entire switch will need to be replaced. Both types are supported for dual five-minute vault periods by the SPS units, and are mounted in the same frame allowing them to use the same rack rail kits for mounting in the bay.

## Bay Components - Engine

### Engine

Contains two every directors

- Even (top)
- Odd (bottom)
- Redundant power to directors

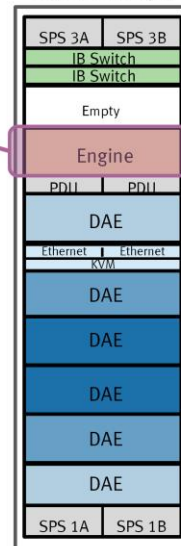
Front of Cabinet



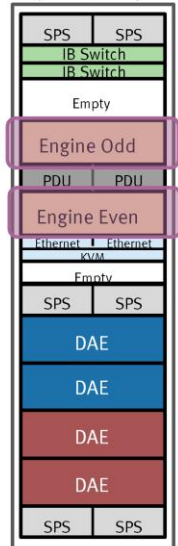
Rear of Cabinet



Single Engine System Bay 1



Dual Engine System Bay 1



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

18

Each engine in the VMAX<sup>3</sup> Family of arrays contains two directors. The Even Director in a given engine is located on the top, and the Odd Director is on the bottom as you are looking at the engine. Each director within an engine has redundant power with dual power connections to each director.

## Bay Components – Power Distribution Unit (PDU)

### PDU

Two power zones

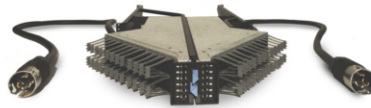
- Zone A – gray cables
- Zone B – black cables

Power options

- Three-Phase delta (two drops/bay)
- Three-Phase wye (two drops/bay)
- Single-Phase (up to six drops/bay)

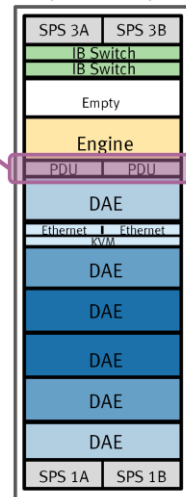


Front of Cabinet

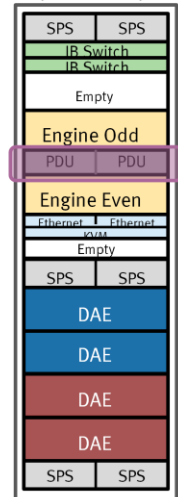


Rear of Cabinet

### Single Engine System Bay 1



### Dual Engine System Bay 1



EMC<sup>2</sup>

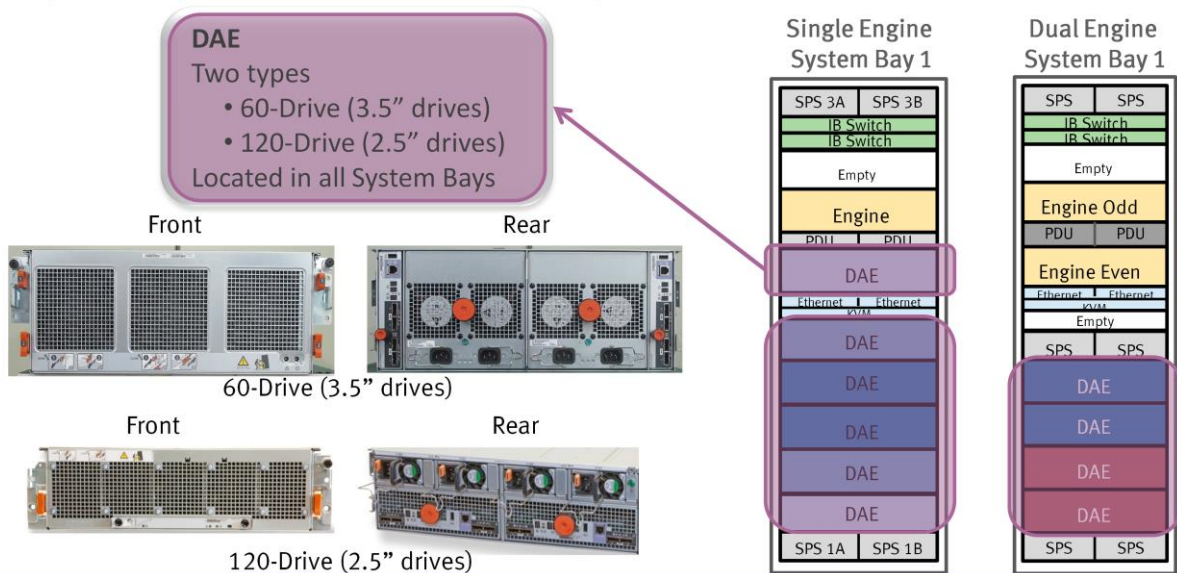
Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

19

The Power Distribution Unit (PDU) combines the previous Power Distribution Panel (PDP) and PDU functions. Two 24-outlet PDU units, one per power zone (Zones A and B), are used in the VMAX<sup>3</sup> Family of arrays. Breakers are protected in six groups of four, and are accessible through the rear door. Every power cord has a specific outlet in which it belongs - to support failure modes and loading on phases. PDUs are available in Three-Phase Delta, Three-Phase Wye and Single-Phase power configurations. Wiring diagrams and figures are available in Simplified Symmwin and product manuals.

## Bay Components – Disk Array Enclosure (DAE)



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

20

Two types of DAEs are available and located in all system bays in the VMAX<sup>3</sup> Family arrays. A 60-drive DAE holds up to sixty 3.5" disk drives in various sizes and speeds, and a 120-drive DAE holds up to (120) 2.5" drives, also available in multiple sizes and speeds. More details on DAEs and drives will follow in an upcoming module.

## Bay Components – Ethernet Switch

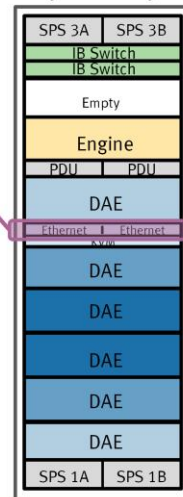
### Ethernet Switch

System Bay 1 only (two switches)  
1 Gbps/port  
Carries environmental information  
Connects to each engine and Fabric

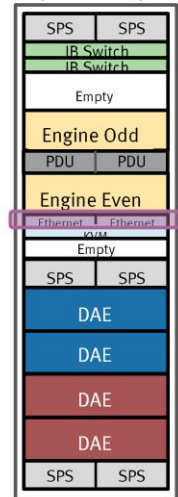


Rear of Cabinet

### Single Engine System Bay 1



### Dual Engine System Bay 1



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

21

A pair of Ethernet switches located in System Bay 1 of both single engine and dual engine bays provides connectivity to each engine and the Fabric for environmental communications. Each switch is a 16-port, unmanaged switch with a single power supply. Cabling to the switches is port-specific, and wiring diagrams are available in Simplified Symmwin and product manuals. Ethernet A cables are green, Ethernet B cables are violet.

## Bay Components – Keyboard/Video/Mouse (KVM)

**KVM**  
System Bay 1 only  
Service Access  
• Simplified Symmwin

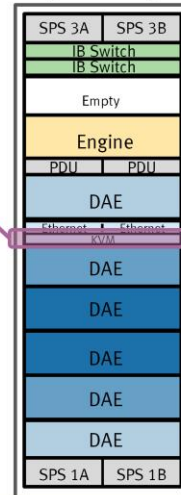


Front

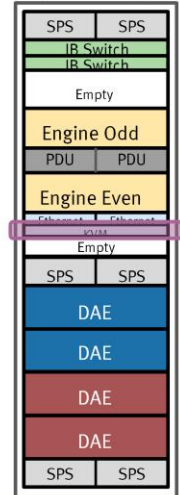


Rear

Single Engine  
System Bay 1



Dual Engine  
System Bay 1



**EMC<sup>2</sup>**

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

22

The Keyboard, Video, Mouse (KVM) component is located in System Bay 1 of single engine and dual engine systems. The KVM provides service access to Simplified Symmwin for onsite service activities.

## Bay Components – Work Tray

### Work Tray

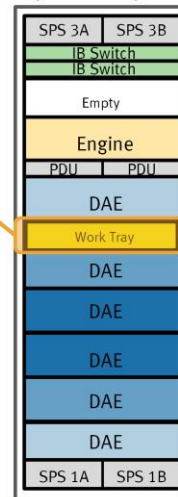
System Bays 2-8

Point-of-service tray for laptop

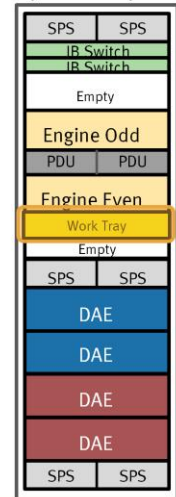
- includes power cable
- includes Ethernet cable



Single Engine  
System Bay 2-8



Dual Engine  
System Bay 2-4



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

23

A Work Tray is located in Bays 2 through 8 in the VMAX<sup>3</sup> Family arrays. The Work Tray provides connectivity for a laptop for remote access to scripted procedures from any bay other than System Bay 1. This can be especially helpful when servicing bays in a large data center or on a dispersed system.

# Module 2: Hardware Components

## Lesson 1: Summary

During this lesson the following topics were covered:

- Single Engine Bay Components
- Dual Engine Bay Components

This lesson covered system bay component locations.

# Module 2: Hardware Components

## Lesson 2: VMAX<sup>3</sup> Engine

This lesson covers the following topics:

- Engine Components



This lesson covers VMAX<sup>3</sup> Engine components.

# VMAX<sup>3</sup> Engine

## Common Features

### VMAX 100K, 200K, 400K

4U Enclosure

Supports up to (6) DAEs

2 Director Boards

- Even director - top
- Odd director - bottom
- Each with redundant power and cooling

Management

- Two Management Module Control Stations (MMCS) - Engine 1
- Two Management Modules (MMs) - Engines 2-8

11 I/O Slots per Director

- Two Back-end I/O Modules (Slots 4, 5)
- One Fabric Connect (SIB) (Slot 10)
- Four Vault to Flash (Slots 0, 1, 6, 7)
- Four Universal Slots for Front end (Slots 2, 3, 8, 9)

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

26

All VMAX<sup>3</sup> with HYPERMAX OS 5977 engines are 4U enclosures, supporting up to six DAEs for a maximum of (360) 3.5" or (720) 2.5" drives. The engines contain two director boards, each board with redundant power and cooling. Engine 1 contains the Integrated Service Processor in the form of Management Module Control Stations (MMCS) in each director. In all arrays, there is a primary and secondary MMCS in Engine 1 for continuous access in the event of a failure. All other engines contain standard Management Modules (MMs). Eleven I/O slots per director are available: two back-end, one fabric connect, four Flash used for vaulting, and four universal slots for front-end modules.

## Engine Types

### VMAX 100K

- 6-core, 2.1 GHz Ivy Bridge CPU
- 512GB or 1TB Cache/Engine

### VMAX 200K

- 8-core, 2.6 GHz Ivy Bridge CPU
- 512GB, 1TB or 2TB Cache/Engine

### VMAX 400K

- 12-core, 2.7 GHz Ivy Bridge CPU
- 512GB, 1TB or 2TB Cache/Engine



Front View



Rear View

EMC<sup>2</sup>

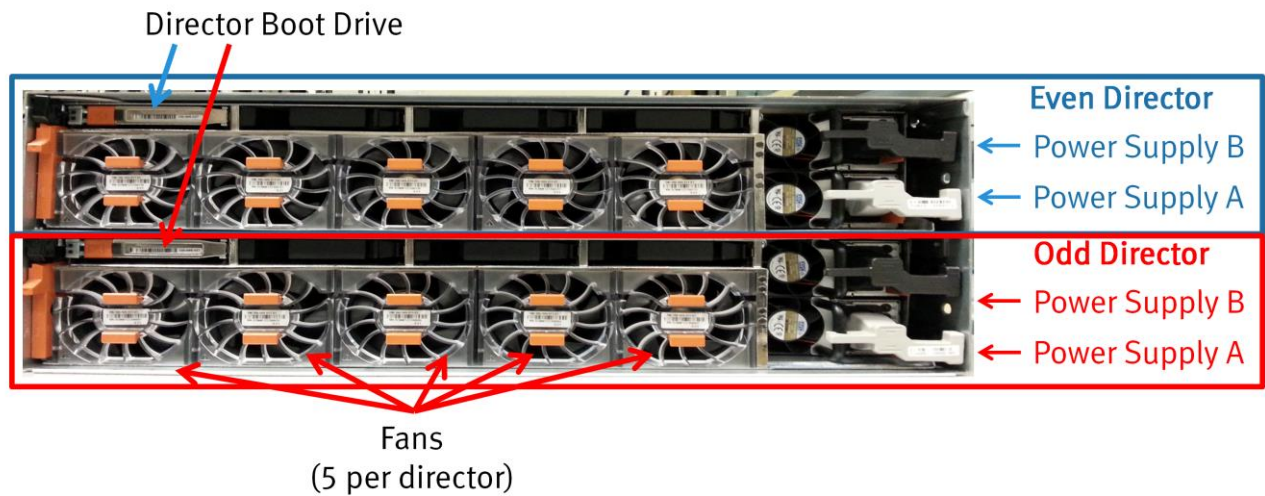
Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

27

Three types of engines are available with the VMAX<sup>3</sup> Family of arrays. The VMAX 100K uses 6-core, 2.1 GHz Ivy Bridge CPUs in each director and cache options of 512 Gigabytes or 1 Terabyte per engine. For VMAX 200K, 8-core 2.6 GHz Ivy Bridge CPUs are used, and 512 Gigabytes, 1 Terabyte or 2 Terabyte cache per engine is available. The VMAX 400K has 12-core 2.7 GHz Ivy Bridge CPUs, and like the 200K, has 512 Gigabytes, 1 Terabyte or 2 Terabyte cache options per engine. In all engines, Hyperthreading is used, logically doubling the amount of cores.

## Engine Front



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

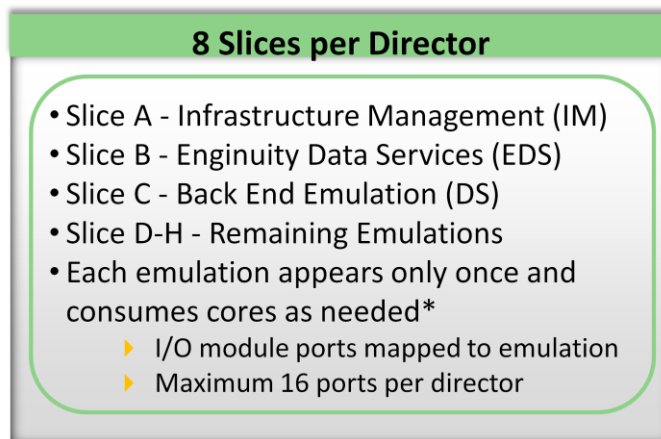
VMAX3 Hardware Fundamentals

28

There are two directors per engine: an Even Director (for example, Director 2, 4, 6 or 8) and an Odd Director (such as Director 1, 3, 5 or 7). A boot drive\* is included in each director, along with five cooling fans and redundant power supplies. Power Supply A is on the bottom of each director and uses a gray connector. Power Supply B is on the top, using a black connector.

\*The boot drive is used to boot HYPERMAX OS 5977 onto the director. It holds emulation binaries and contains firmware binaries for the devices in the system. It also contains the director's error logs and some IMPL related configuration data. This is what the EEPROM was in legacy systems. In legacy systems, the boot device was installed on the director board and not removable, unless the director was removed from the chassis. In this system, the 16GB boot drive is accessible and can be replaced while the director is online.

# Director Emulations



\*As emulations demand, more (or fewer) cores are assigned

In the VMAX<sup>3</sup> Family of arrays, there are eight slices per director. Slice A is used for the Infrastructure Manager (IM) system emulation. The goal of the IM emulation is to place common infrastructure tasks on a separate instance so that it can have its own CPU resources.

The IM performs all of the environmental monitoring and servicing. All environmental commands, syscalls and FRU monitoring are issued on the IM emulation only. DAE FRUs are monitored by the IM through the DS emulation. If the DS emulation is down, access to DAE FRUs is affected.

Slice B is used by Enginuity Data Services (EDS) system emulation. EDS consolidates various Enginuity functionalities “under one roof” to allow easier and more scalable addition of features. It provides a common “host” for data services, which are not directly in the data path, such as FAST processing, write misses, destaging data and allocation of cache space.

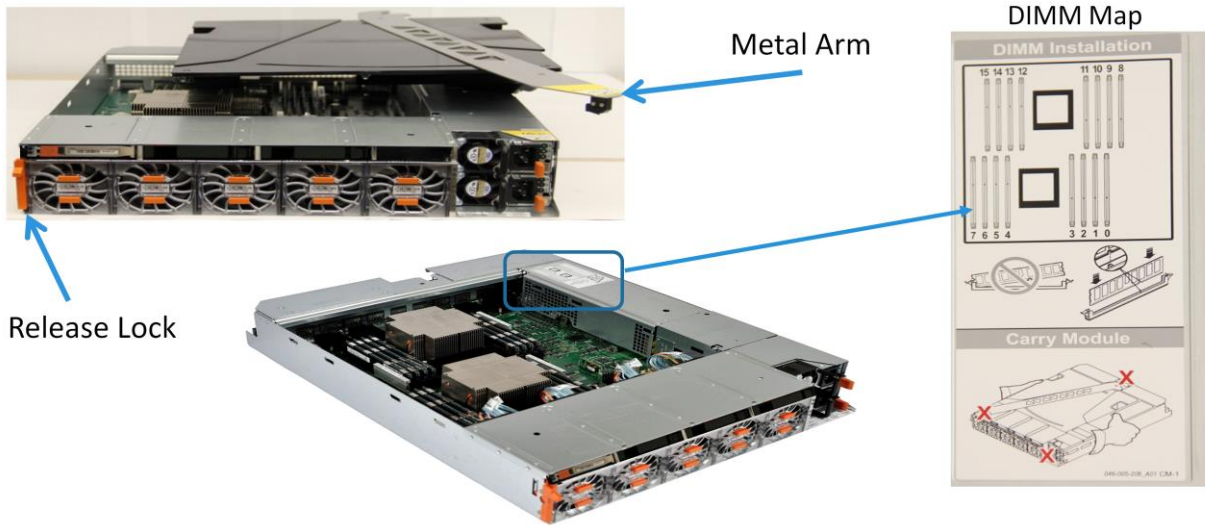
Slice C is used for back end emulation (DS).

Slices D through H are used for the remaining emulations. At GA, Fibre Channel (FA) is the only supported front end emulation, therefore, Slice D is currently used for FA emulation. Slices E through H are reserved for future use.

Each emulation appears only once per director and consumes cores as needed. A maximum of 16 front end I/O module ports are mapped to an emulation. In order for a

front end port to be active, it must be mapped to an emulation.

## Director and DIMMS



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

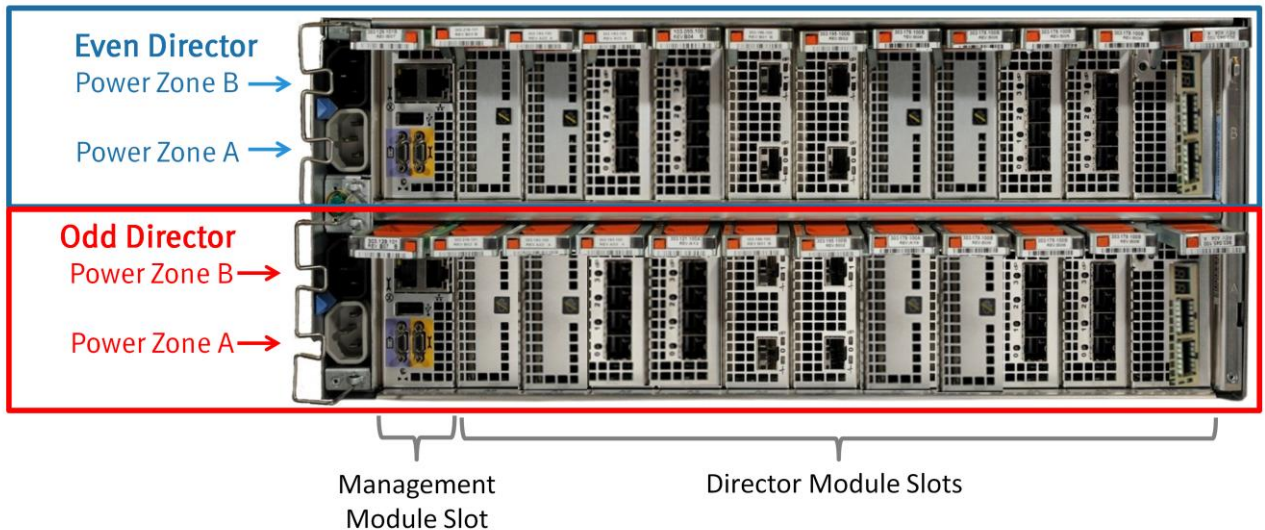
30

Directors are removed through the front of the engine. An orange release lock is pressed down, and the metal arm is moved from left to right to assist in the removal of the director. A black plastic protective cover is popped off the top of the director to reveal DIMM FRUs.

Each director contains (16) DIMMs:

- VMAX 400K = 16, 32, 64GB DIMMS
- VMAX 200K = 16, 32, 64GB DIMMs
- VMAX 100K = 16, 32GB DIMMs

## Engine Rear



The rear of the engine has redundant power input to both directors, and contains slots for both Management and Director Modules. Module configurations will be discussed in upcoming slides.

# Module 2: Hardware Components

## Lesson 2: Summary

During this lesson the following topics were covered:

- Engine Components



This lesson covered VMAX<sup>3</sup> Engine components.

## Module 2: Hardware Components

### Lesson 3: Disk Array Enclosures (DAE)

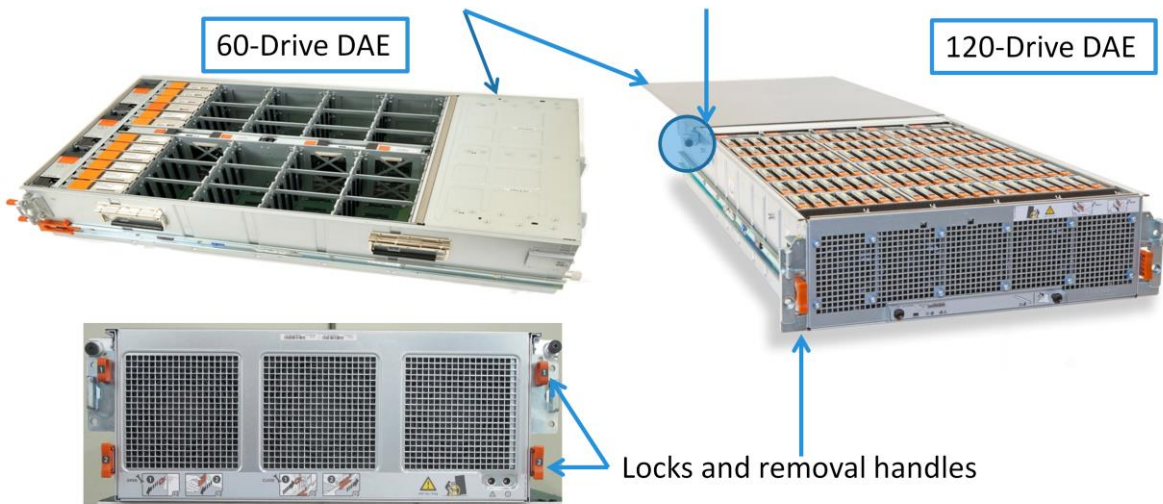
This lesson covers the following topics:

- Disk Array Enclosures (DAEs)

This lesson covers DAEs for the VMAX<sup>3</sup> Family arrays.

## Disk Array Enclosures (DAE) Features

Top cover attached to the rack with the two captive screws



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

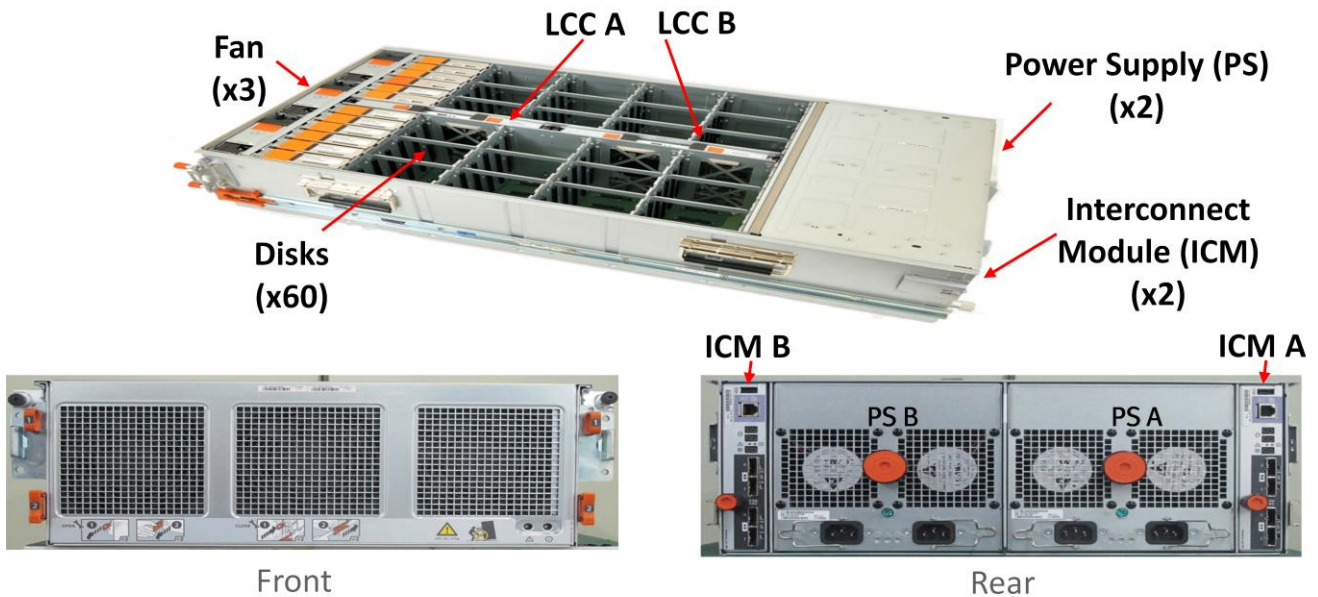
VMAX3 Hardware Fundamentals

34

Two types of DAEs are available with the VMAX<sup>3</sup> family of arrays: a 60-drive DAE, and a 120-drive DAE. Both DAEs have a top cover attached to the rack with two captive screws. This cover provides protection and environmental functionality (proper air flow and dust prevention) when the DAE is in the system. When sliding a DAE out of the chassis, the cover should remain inside the system. Therefore, do not loosen the captive screws to release the DAE, but rather use the orange locks and removal handles on the front of the DAEs to slide it out of the system to access DAE components. When servicing components in a DAE, an interlock system prevents pulling more than one DAE out of the cabinet for safety reasons. Please note: When third party racking is used, the interlocking system is not in place, and care must be taken to ensure that only a single DAE is pulled out of the system at a time to prevent the cabinet from tipping due to improper weight distribution. Although the DAE can remain out of the system during service activities, it should be slid back in if not servicing to maintain proper cooling.

Filler panels, or drive blanks (not shown in this display) should be used in slots with no drives installed to maintain proper air flow.

## 60-Drive DAE Components



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

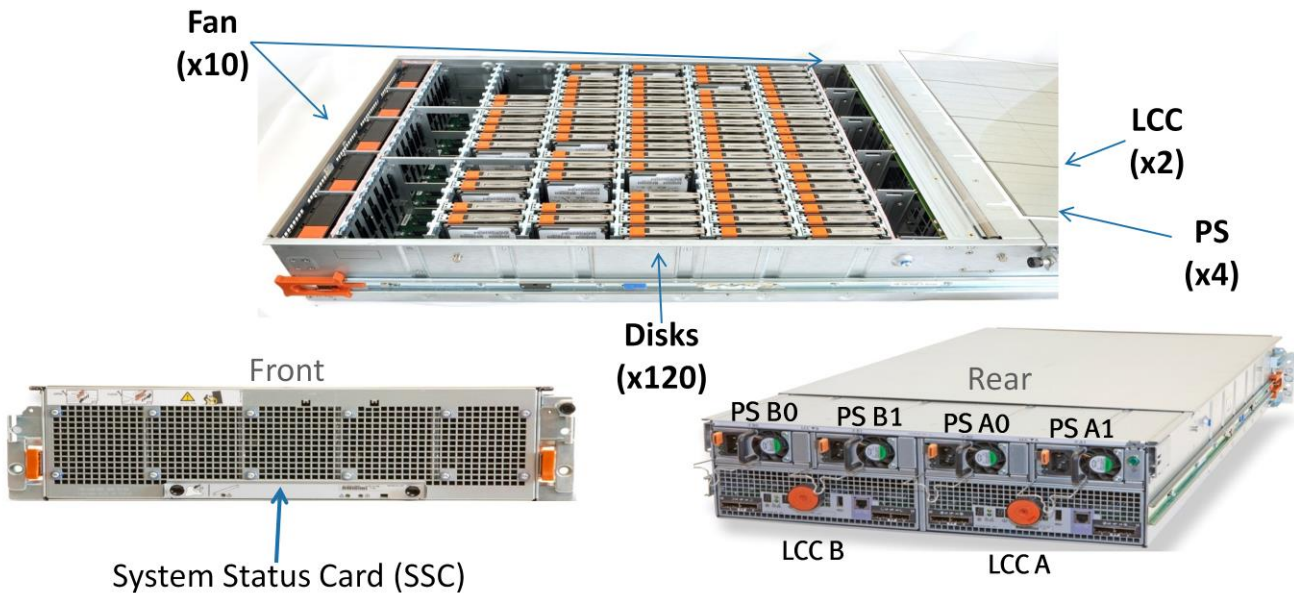
VMAX3 Hardware Fundamentals

35

60-Drive DAE components include:

- Fans = N+1 (three total) with adaptive cooling (if one fails, others will speed up to ensure proper cooling)
- LCC = Link Control Card - connectivity to disks
- Disks = 60 Disks - 3.5", multiple sizes and speeds
- PS = Power Supply - PSA and PSB for redundant power (each with dual power connections from PDU)
- ICM = Inter Connect Module (back-end modules connect to LCCs through the cable connections to the ICM)

## 120-Drive DAE



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

36

120-Drive DAE components include:

- SSC = System Status Card - Responsible for environmental monitoring of the DAE components
- Fans = N+2 (10 total) with adaptive cooling (if one fails, others will speed up to ensure proper cooling)
- LCC = Link Control Card - connectivity to disks
- Disks = 120 Disks - 2.5", multiple sizes and speeds
- PS = Power Supply - PSA0/PSA1 and PSB0/PSB1 for redundant power

# Module 2: Hardware Components

## Lesson 3: Summary

During this lesson the following topics were covered:

- Disk Array Enclosures (DAEs)



This lesson covered DAEs for the VMAX<sup>3</sup> Family arrays.

## Module 2: Summary

Key points covered in this module:

- Identify hardware components

This module covered the VMAX<sup>3</sup> with HYPERMAX OS 5977 hardware components locations and functionality.

## Module 3: Configuration Options

Upon completion of this module, you should be able to:

- Discuss configuration options

This module focuses on configuration of the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

# Module 3: Configuration Options

## Lesson 1: System Configurations

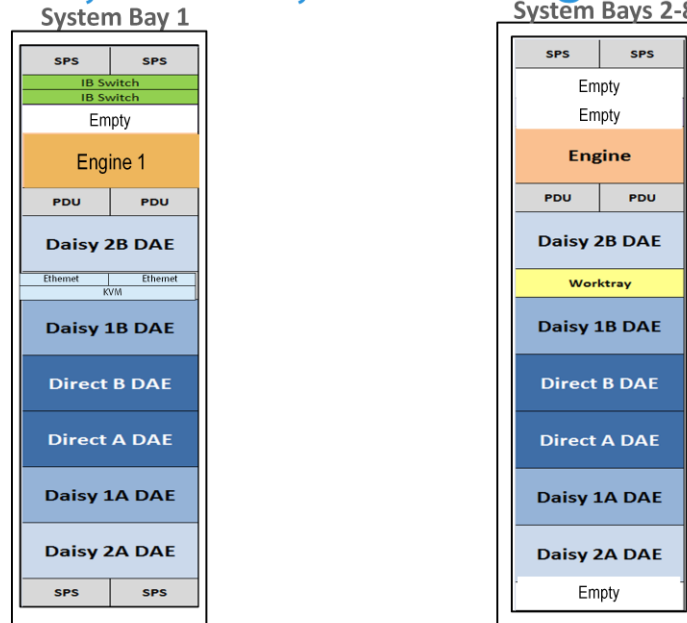
This lesson covers the following topics:

- Bay Configurations



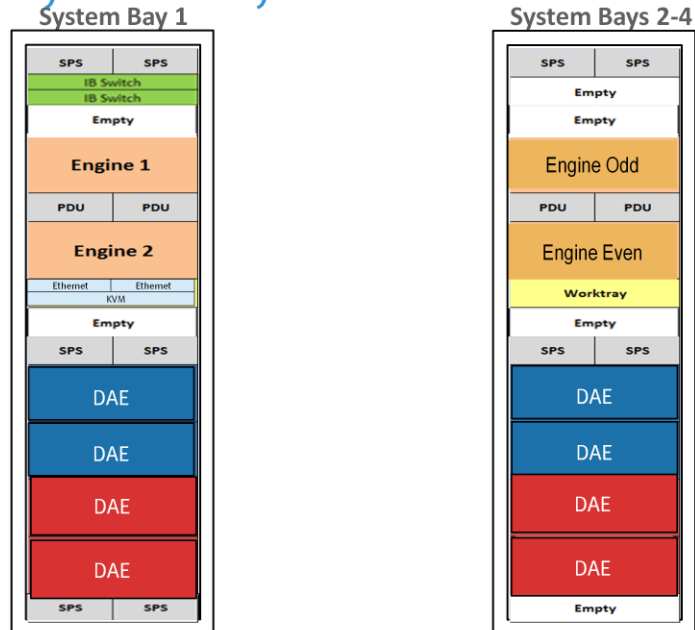
This lesson covers bay configuration options for the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

# Single Engine System Bay DAE Configuration



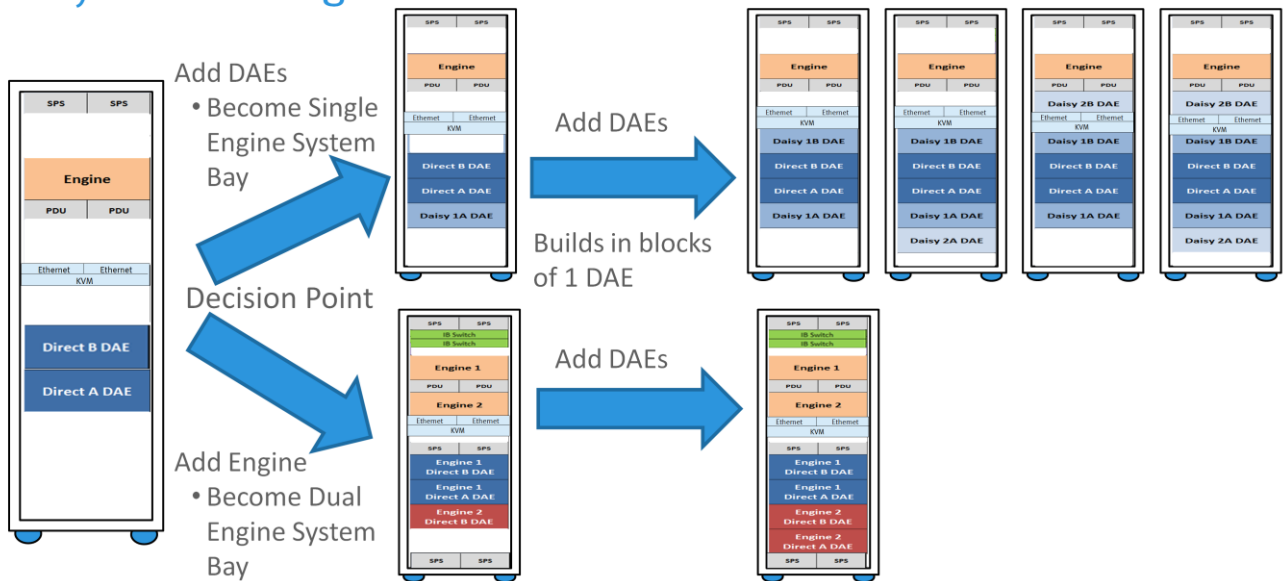
Single engine system bays, as the name describes, have one engine per bay. The engine is battery supported by a pair of SPS units in a single tray along the top of the cabinet. Single engine bays contain a maximum of six DAEs for disk connectivity to the engine. The engine supports two chains of drives (labeled as Direct A and Direct B in the diagram), each with a direct-attached DAE, and potentially one or two daisy-chained DAEs. A dual engine system bay is also offered, shown on the next slide. A single system cannot include both single engine and dual engine bays; all bays are either single engine or dual engine.

# Dual Engine System Bay



Dual engine system bays contain either one or two engines, and a maximum of four DAEs. Each engine is battery supported by a pair of SPS units (SPS 1A/1B and SPS 3A/3B), and two chains of direct-attached drives (labeled as Direct A and Direct B per engine in the diagram). As mentioned in the previous slide, single engine and dual engine bays can not be mixed in the same array.

# System Configuration



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

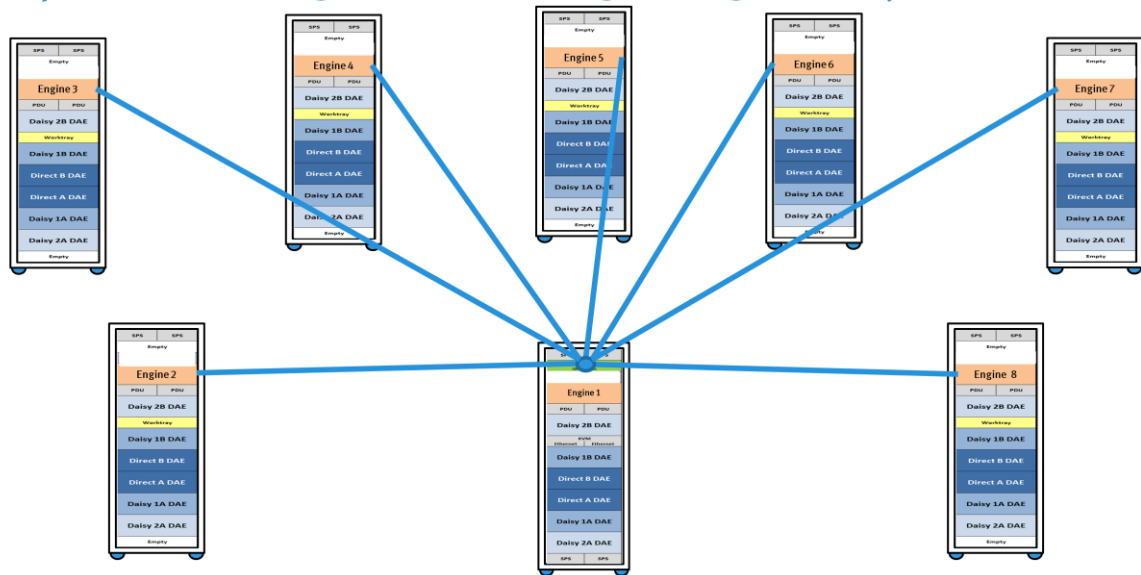
VMAX3 Hardware Fundamentals

43

When a single engine system is ordered with one or two DAEs, the upgrade path is open. If you choose to add daisy-chained DAEs, the system permanently becomes a single engine bay.

Additional DAEs can be added in single increments for increased capacity. If you choose to add an engine and supporting DAEs, the system becomes a dual engine bay and direct-attached DAEs can be added to support the second engine.

## Dispersed Configuration - Single Engine Bays



EMC<sup>2</sup>

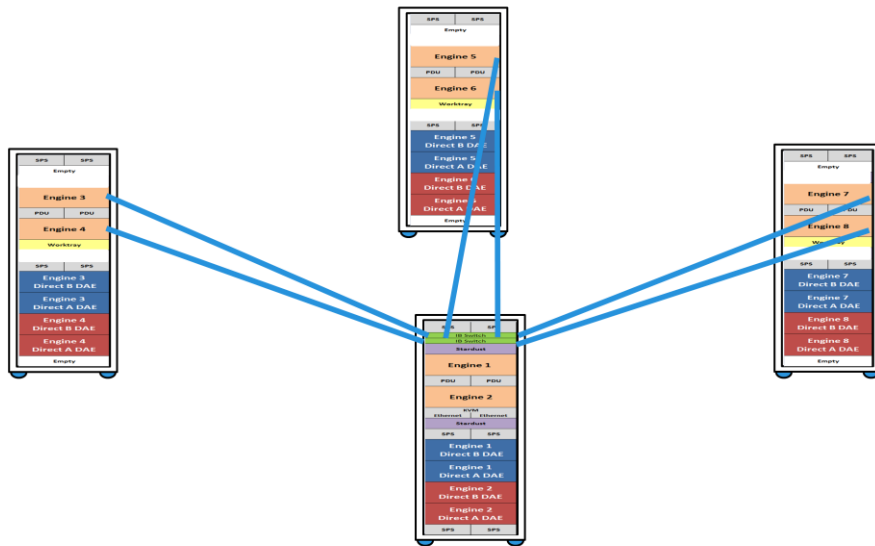
Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

44

Dispersion of System Bays 2 through 8 is supported in the VMAX<sup>3</sup> Family of arrays. The bays can be dispersed up to 25 meters from System Bay 1 to allow for flexibility in configurations. Blue lines in the picture above represent multiple Infiniband and Ethernet links.

## Dispersed Configuration - Dual Engine Bays



Dual engine configurations also support dispersion of bays 2 through 4 up to 25 meters from System Bay 1.

## Third Party Racking

- Flexibility for standardization of data centers using standard customer racks
- NEMA standard 19" mounting racks
- Empty customer racks providing enough rack-to-rack pass-through cable access



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

46

VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays can be placed in a third party rack to provide customers flexibility in designing their datacenters. Standard 19" NEMA mounting facilities must be used, and the rack needs to be empty – no other components can occupy the rack space designated for the system. For additional details and installation requirements for third party racking, see the VMAX Family with HYPERMAX OS 5977 Third Party Racking Course.

# Module 3: Configuration Options

## Lesson 1: Summary

During this lesson the following topics were covered:

- Bay Configurations



This lesson covered bay configuration options for the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

# Module 3: Configuration Options

## Lesson 2: Drive and DAE Configurations

This lesson covers the following topics:

- DAE Types
- Supported Drives
- Power Zoning
- DAE Mixing
- Sparing



This lesson covers DAE and drive configuration options for the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

## 60-Drive DAE

- 6G SAS connectivity
- (60) 2.5" or 3.5" drives
- Redundant power
- N+1 Fans with adaptive cooling
- 225 lbs ( 102 kg) fully populated



### 2.5" Drives (in 3.5" carrier)

200, 400, 800GB EFD

300GB/15K RPM

300, 600GB/10K RPM

1.2TB/10K RPM

### 3.5" Drives

2, 4TB/7.2K RPM

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

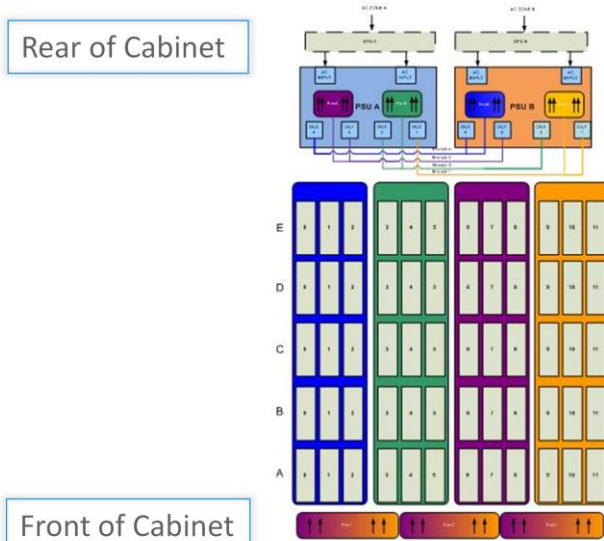
VMAX3 Hardware Fundamentals

49

The 60-Drive DAEs use 6 Gb SAS connectivity to connect to up to 60 drives. Redundant power, and N+1 fan configuration is provided for high availability. When fully populated, the DAE weighs 225 lbs (102 kg), and therefore, a lift tool is needed for replacements.

Drives currently available in the 60-Drive DAE include 200, 400 and 800GB EFD; 300GB 15K RPM; 300GB, 600GB and 1.2TB 10K RPM, and 2 and 4TB 7.2K RPM drives. Always check release notes and [emc.support.com](http://emc.support.com) for the latest supported drives.

## 60-Drive DAE Power Zones



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

50

In order to have isolation of RAID group members for High Availability (HA), DAEs have been divided into four power zones. This allows RAID groups to be striped across the four distinct power zones. Most failures will only impact one RAID member.

A minimum of 1 DAE is required for Raid 1, Raid 5 3+1 and Raid 6 6+2.

A minimum of 2 DAEs is required for Raid 5 7+1 and Raid 6 14+2.

## 120-Drive DAE

- 6G SAS connectivity
- (120) 2.5" drives
- N+1 power
- N+2 Fans with adaptive cooling
- 150 lbs (68 kg) fully populated



Rear

### 2.5" Drives

200, 400, 800GB EFD

300GB/15K RPM

300, 600GB/10K RPM

1.2TB/10K RPM



Front

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

51

The 120-Drive DAEs use 6 Gb SAS connectivity to connect to up to 120 drives. N+1 power and N+2 fan configuration is provided for high availability. When fully populated, the DAE weighs 150 lbs (68 kg), and therefore, a lift tool is needed for replacements.

Drives currently available in the 120-Drive DAE include: 200, 400 and 800GB EFD; 300GB 15K RPM; 300GB, 600GB and 1.2TB 10K RPM drives. Always check release notes and [support.emc.com](http://support.emc.com) for the latest supported drives.

# 120-Drive DAE Power Zones



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

52

As with the 60-drive DAE, in order to have isolation of RAID group members for High Availability (HA), DAEs have been divided into four power zones in the 120-drive DAE. Again, this allows RAID groups to be striped across the four distinct power zones, and most failures will only impact one RAID member.

A minimum of 1 DAE is required for Raid 1, Raid 5 3+1 and Raid 6 6+2.

A minimum of 2 DAEs is required for Raid 5 7+1 and Raid 6 14+2.

## DAE Mixing

- Behind an engine mixing in increments of one DAE
  - ▶ Example: Engines with direct-attached 60-drive DAE connected to a daisy-chain 120-drive DAE
- DAEs balanced across engines



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

53

DAE mixing is supported in the VMAX<sup>3</sup> Family of arrays. As shown here, a dual engine system can have both types of DAEs installed, mixing behind a single engine. For example, a 60-drive direct-attached DAE can have a daisy-chained 120-drive DAE.

In new system configurations DAEs are populated across all of the engines in that system. Users can not individually specify DAE type and location while configuring in the DX/CX configurator. The Sizing and configuration tools will apply the required drive types and drive mixes as evenly as possible across all engines in the system using the population sequence rules below:

Symmwin manages population of components

# 2.5" drives / 120 = # 120-Drive DAEs (rounded up to the next whole number)

# 3.5" drives / 60 = # of 60-Drive DAEs (rounded up to the next whole number)

All 2.5" 120-Drive DAE Mod 8 (RAID5 7+1, RAID6 14+2) populated 1<sup>st</sup> striped in pairs across available engines until consumed

All 2.5" 120-Drive DAE Mod 4 (RAID1, RAID5 3+1, RAID6 6+2) populated 2<sup>nd</sup> striped in single DAEs across available engines until consumed

All 3.5" 60-Drive DAE Mod 8 populated 3<sup>rd</sup> striped in pairs across available engines until consumed

All 3.5" 60-Drive DAE Mod 4 populated 4<sup>th</sup> striped in single DAE across available engines until consumed

Based on this population, the .dll will return # of Direct 120-Drive, Daisy 120-Drive, and Direct 60-Drive, and Daisy 60-Drive DAEs to DX/CX

DAEs will be balanced across engines

## Drive Sparing

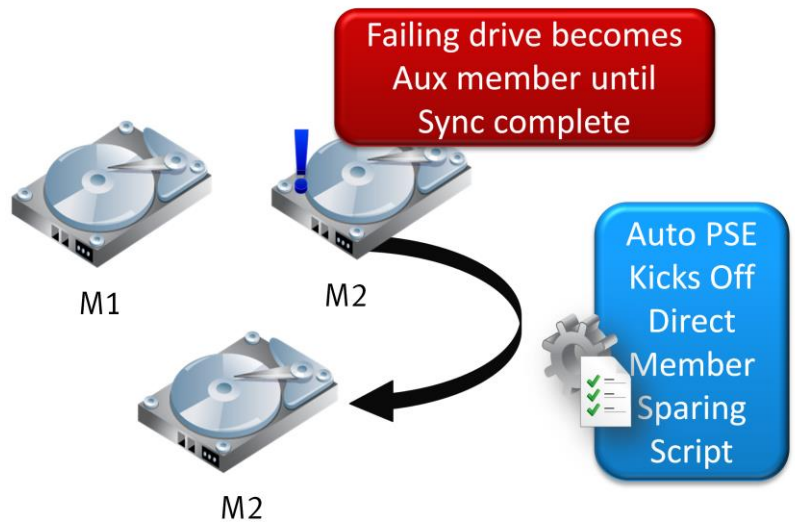
- Spares are configured as empty disks
  - ▶ Each disk group must have a minimum of one spare configured
  - ▶ 1 spare minimum per 50 disks
  - ▶ Spares cannot be shared between disk groups
  - ▶ Same rules for HDD and Flash



If the disk group spans multiple engines each engine must have a spare because local RAID is local to one engine.

## Direct Member Sparing (DMS)

- Default process for AutoPSE
- Support now includes RAID 6 (14+2)
- Spare can be moved to another DA within the same engine



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

55

This slide shows the Direct Member Sparing Process. When a drive fails, the AutoPSE runs the Direct Member Sparing script. The DMS procedure adds the new drive, BUT will keep the failing drive, which becomes defective, but "Ready." Keeping the new drive in a Ready state, we can now copy data from that failing drive. This becomes very beneficial with Parity RAID protection, such as RAID 5, instead of rebuilding data from all the drives in the RAID Group, we can now copy directly from the failing drive, where possible. Any remaining tracks unable to copy will be rebuilt from the remaining members. This cuts rebuilding time significantly. The failing drive is removed only at the end of the rebuild.

# Module 3: Configuration Options

## Lesson 2: Summary

During this lesson the following topics were covered:

- DAE Types
- Supported Drives
- Power Zoning
- DAE Mixing
- Sparing



This lesson covered DAE and drive configuration options for the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

# Module 3: Configuration Options

## Lesson 3: Director Configurations

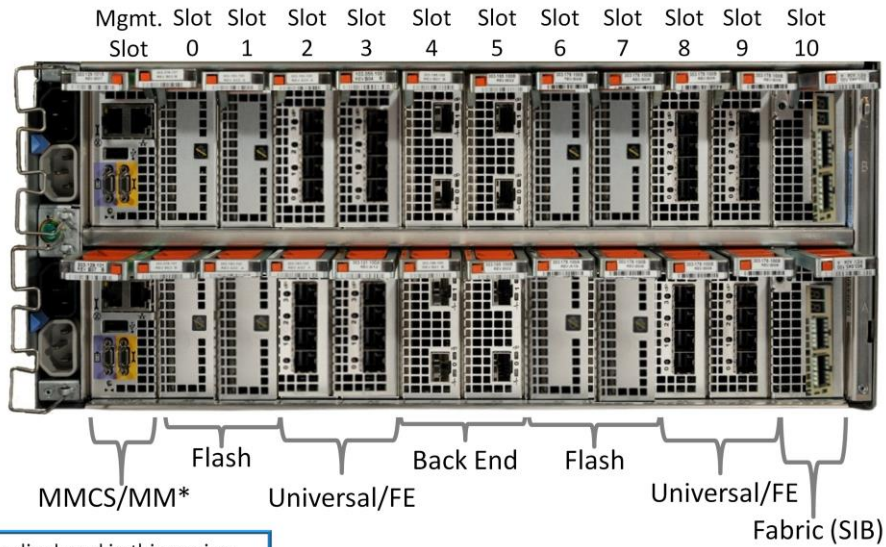
This lesson covers the following topics:

- Director Module Configurations
- Port Configurations



This lesson covers director module configuration options for the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

## Engine Rear



\*Management Module displayed in this engine

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

58

The Management slots in the rear of the engine contain the Management Module Control Stations (Engine 1 only) or the Management Modules (all other Engines), Flash modules, Back-end and Front-end I/O modules, and Fabric-connect modules (SIBs). Slots are numbered 0 through 10 across the top, and module types are slot-specific, as labeled.

## Management

### Management Module Control Station (MMCS)

Engine 1 only  
Two MMCS - one per director  
Management and monitoring  
Embedded Service Processor

### Management Module (MM)

Engines 2-8  
Two MMs per engine  
Management and monitoring

### MMCS

Embedded MiniSP



MM

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

59

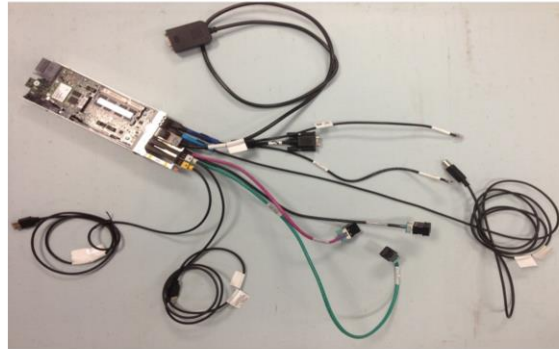
Within each VMAX<sup>3</sup> with HYPERMAX OS 5977 engine, two management modules are used by the directors to monitor and control the environment in which the engine operates. The management modules are responsible for monitoring the SPS units and communicating with other engines in the system. In Engine 1, Management Module Control Stations (MMCS) are used. MMCSs have an embedded “MiniSP” used as the service access or service processor for the system. In Engines 2 through 8, when present, standard Management Modules (MMs) are used for monitoring and communications. The Management Modules do not have the embedded “MiniSP”.

# Management Module Control Station (MMCS)

- Two MMCS per system (Engine 1 only)

**Primary - MMCS-1**  
Installed on Director 1

**Secondary - MMCS-2**  
Installed on Director 2



## Cabling

Color-coded and labeled for identification  
Retention brackets to support cables

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

60

The Management Module Control Station (MMCS) is a replacement for the traditional service processor. This module has management and monitoring capabilities, but also has a “mini SP” embedded to provide service processor functionality.

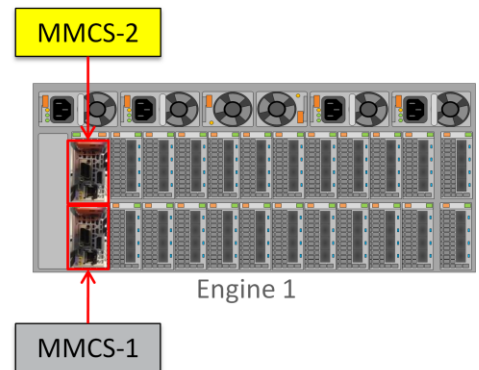
There are two MMCS per system located in management slots on Directors 1 and 2. MMCS-1, located on Director 1, is the Primary and MMCS-2, on Director 2, is the secondary.

The cabling to the MMCS is color-coded and labeled for easy identification, and includes retention brackets to support the cables.

# MMCS Redundancy

## Dual MMCS

- Provides Call Home redundancy
- Each MMCS should have a unique IP address
- Elevated secondary allows support connectivity for replacement of failed:
  - MMCS-1
  - Director 1
  - DIMM on Director 1
- RSync copies all tools and utilities to “other” MMCS
  - Primary to Secondary
  - Elevated Secondary to replaced Primary



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

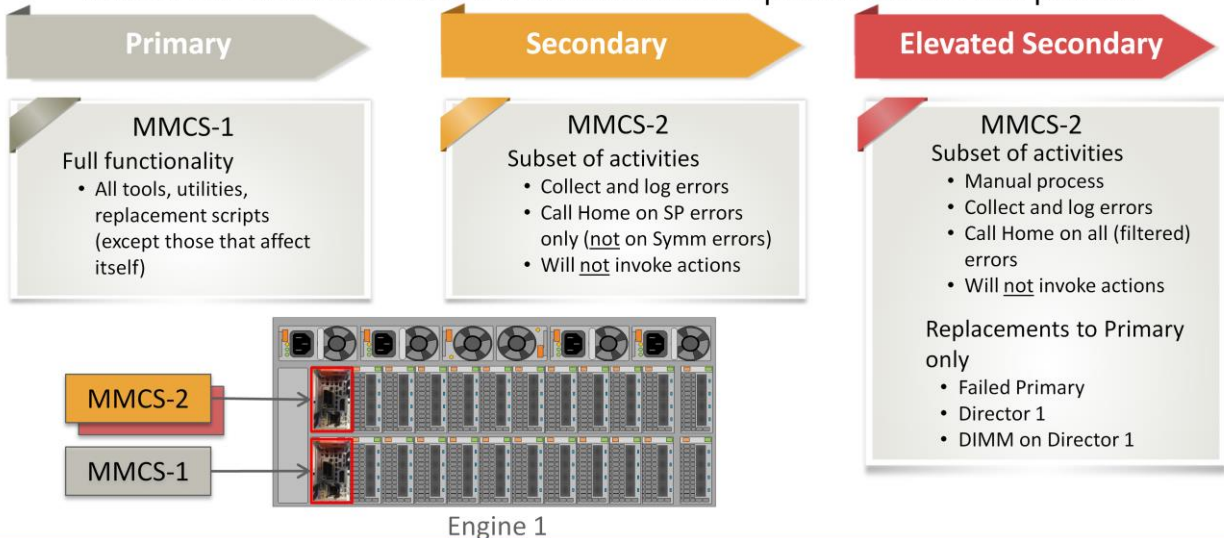
61

In the event of Primary failure, Call home redundancy and replacement activities are performed by the secondary MMCS. A separate IP address is required for MMCS-2 (or a separate modem, if using modem connectivity rather than ESRs), and connecting to it is done using the Red Ethernet cable in the rear of System Bay 1 and a service laptop. Connectivity to MMCS-2 could also be achieved by moving KVM cables from MMCS-1 to MMCS-2, although this is not the preferred method. Detailed instructions are provided during replacement scripts in Simplified Symmwin. Replacement activities available on the Secondary are restricted to those that affect the Primary; that is, Primary MMCS failure, Director 1 failure and Director 1 DIMM failures only. No other maintenance activities can be performed by the Secondary when in an elevated state.

RSync is a utility software that synchronizes files and directories from one location to another. In the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays, RSync is used to synchronize MMCS-1 and MMCS-2.

## MMCS Roles

Role of the MMCS defines the services that a specific MMCS can provide



EMC<sup>2</sup>

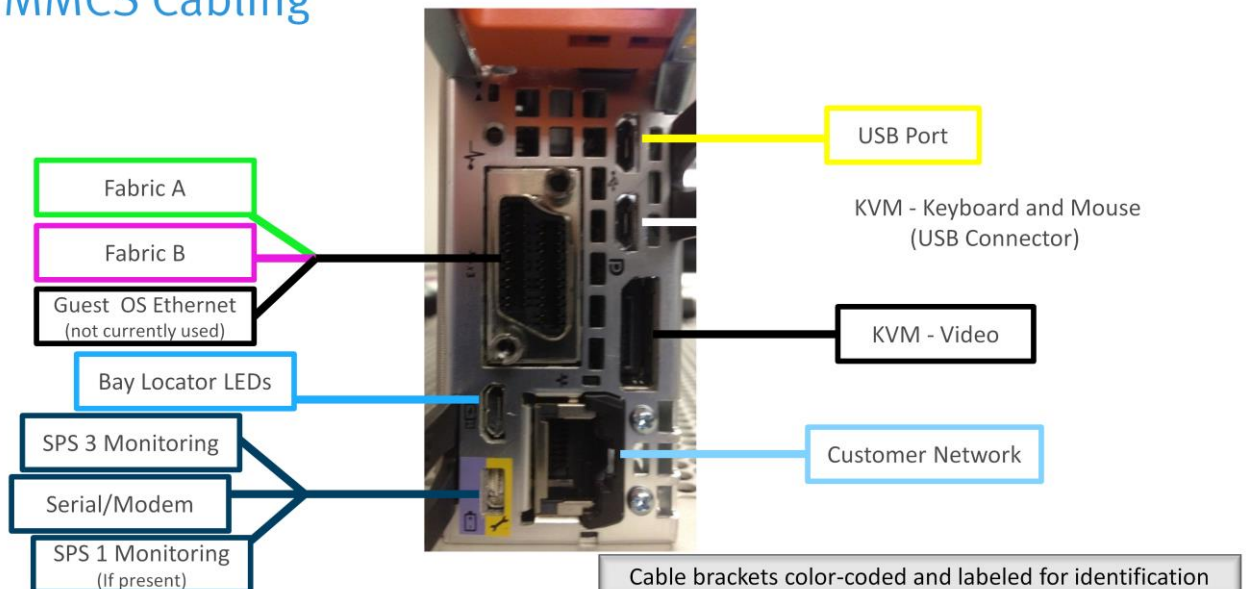
Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

62

The role of the MMCS defines the services that a specific MMCS can provide. The Primary MMCS, MMCS-1, has full functionality. All tools, utilities and replacement scripts (other than those that affect itself) are available. The Secondary MMCS, MMCS-2, provides a subset of activities, such as collecting and logging errors and calling home on SP errors. When a failure occurs on MMCS-1, MMCS-2 is manually promoted to an Elevated Secondary, providing replacement activities for MMCS-1, Director 1, or DIMMs on Director 1 only. All other replacement functionality is not available on the Elevated Secondary. Once the MMCS-1 is promoted back to Primary, all other necessary replacements can be run from it.

## MMCS Cabling



EMC<sup>2</sup>

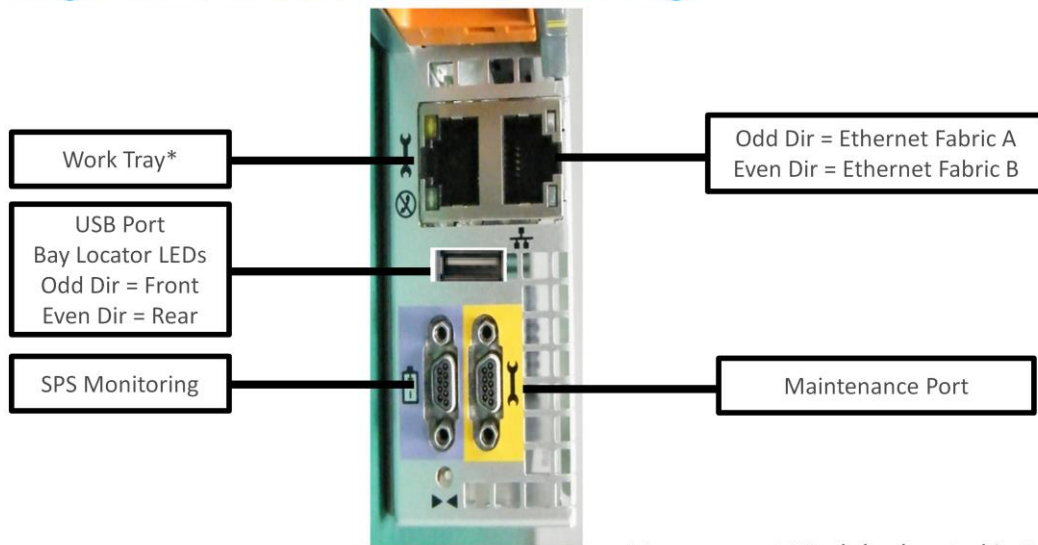
Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

63

Although there are many cables connected to the MMCS for various activities, care has been taken to ensure proper cabling. Color-coding is used to identify cable connections. Retention brackets (also color coded) assist in routing and stabilizing cables once connected. The MMCS is connected to the Ethernet fabrics, Bay Locator LEDs, SPS units for monitoring purposes, KVM, and customer network for ESRS. There is an addition USB connection for service activities and file download/transfer. There is also a connection for a modem, although IP access is recommended and preferred.

## Management Module (MM) Cabling



Management Modules located in Engines 2-8

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

64

The Management Modules (MMs), located in Engines 2 through 8, provide connectivity and monitoring capabilities. Unlike the MMCS, there is no embedded “mini SP”, so most service processor-like functionality is not available. Cabling to the Ethernet switches, Bay Locator LEDs and SPSs are provided in with the MM. A connection for maintenance is also available, but is not used during normal operation. A connection to the Work Tray is provided in Bays 2 through 8 only, for ease-of-access when performing maintenance activities.

## Director Modules – Disk Connectivity

### Back-end I/O Module

Protocol: SAS (6Gb)

Slots: 4 and 5

Ports: 2



Front View



Top View

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

65

The back-end I/O module is used for connectivity to disks. It uses two 6 Gb SAS connections to the DAEs for access to the disk drives. Back-end I/O modules are populated in Slots 4 and 5 of each director. Back end I/O modules are redundant across directors, with each providing connectivity to both direct attached DAEs.

## Director Modules – Host Connectivity

### Front End I/O Module

Protocol: Fibre  
(FA: 2, 4, 8Gb)  
Slots: 2, 3, 8, 9 (L to R)  
Ports: 4

Protocol: Fibre  
(FA: 4, 8, 16Gb\*)  
Slots: 9, 8, 3, 2 (R to L)  
Ports: 4

2/4/8 GB FA  
Front View



2/4/8 GB FA  
Top View



4/8/16 GB FA  
Front View



\*Bifurcated module - populate right to left

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

66

Front-end I/O modules include both a 2/4/8GB four-port Fibre Channel module and a 4/8/16Gb four-port Fibre Channel module. The 2/4/8Gb module is installed from left to right in Slots 2, 3, 8 and 9. The 4/8/16Gb bifurcated I/O module is installed from right to left in Slots 9, 8, 3 and 2.

Bifurcated = Split. Connection to each I/O module is 8 lanes of PCIe; either one connect of all 8 lanes or bifurcated into 2 connections of 4 lanes each.

## Director Modules – Miscellaneous

**Other**

**Flash for Vault**  
Capacity: 350, 700GB  
Slots: 0, 1, 6, 7

**Fabric Connect (SIB)**  
Slot: 10  
Present in multi-engine systems only



Flash  
Front View



Fabric Connect (SIB)  
Front View



Fabric Connect (SIB)  
Top View

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

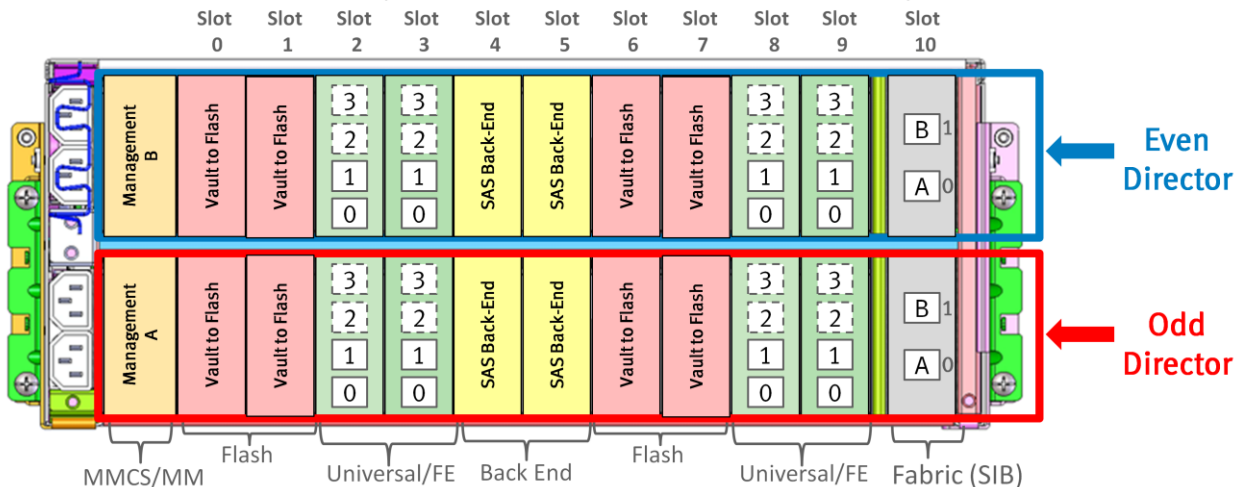
67

Vaulting in the VMAX<sup>3</sup> with HYPERMAX OS 5977 array is done to a Flash module, rather than to drives as in legacy VMAX arrays. The Flash module codenamed “Whirlwind” is available in 350 and 700GB sizes. Flash modules are installed in Slots 0, 1, 6 and 7 in each director. Vaulting to Flash allows the ability to use larger global memory space per director that can be saved within 5 minutes which aids better performance (512GB in 4 mins). This also reduces the extra hardware necessary in previous generations that vaulted to disk, thus saving cost, space (lighter system) and giving more capacity back to the customer.

Connectivity to the Fabric for each director is provided by the SIB module codenamed “Gorgon”. The SIB is installed in Slot 10 in each director in multi-engine systems only.

## Physical Port Layout

- Ports are physically numbered 0-3 from bottom to top
- I/O Modules with 2 ports numbered 0-1, from bottom to top



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

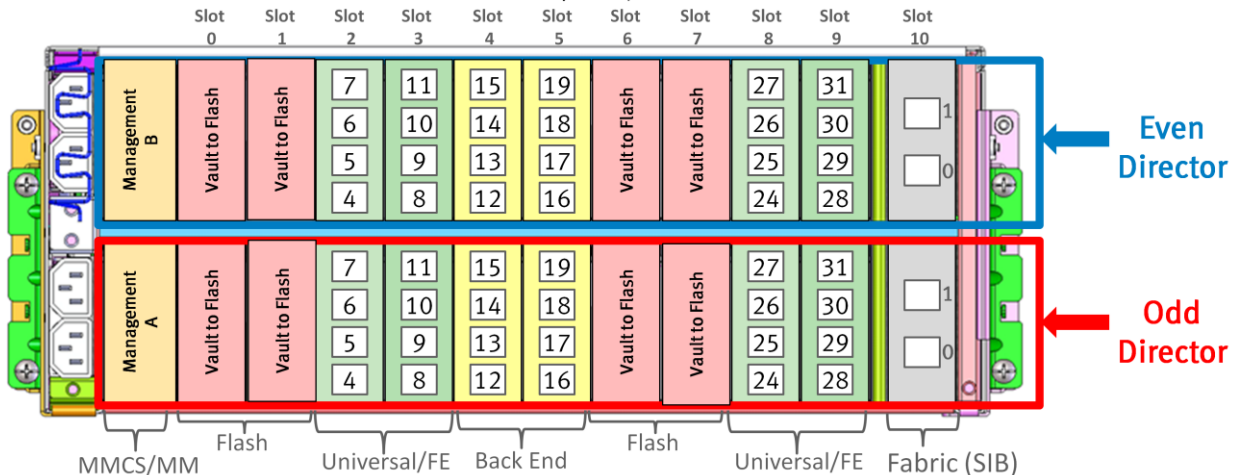
VMAX3 Hardware Fundamentals

68

Physical port numbering has not changed. When connecting to an I/O module, ports are numbered 0 through 3 from top to bottom on the module. I/O modules with two ports (not shown) are labeled Port 0 and 1 from bottom to top. Fabric ports are numbered 0 to 1 from bottom to top as well. Fabric Port 0 connects to Fabric A. Fabric Port 1 connects to Fabric B.

## Logical Port Layout

- Software designed to support 32 ports (Ports 0-31)
- Logical ports numbered left to right, bottom to top across the eight slots used for FE and BE I/O Modules (Note: Ports 0-3 and 20-23 are reserved and not currently used)



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

69

VMAX<sup>3</sup> is designed to support 32 ports per director, ports 0 through 31. These logical ports are numbered left to right, bottom to top, across the eight slots available for front-end and back-end connectivity. Ports 0, 1, 2, 3, 20, 21, 22, and 23 are reserved and not currently used. Slots 4 through 11 and 24 through 31 can be used for front-end connectivity. Ports 12 through 19 are used for back-end connectivity. On the SIB, ports 0 and 1 are used for connectivity to the fabric in each director. Port numbers do not become available unless an I/O module is inserted in the slot.

# Module 3: Configuration Options

## Lesson 3: Summary

During this lesson the following topics were covered:

- Director Module Configurations
- Port Configurations



This lesson covered director module configuration options for the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

## Module 3: Summary

Key points covered in this module:

- Discuss configure options

This module covered the configuration of the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

## Module 4: Accessing the System

Upon completion of this module, you should be able to:

- Access a VMAX<sup>3</sup> array for service

This module focuses on accessing the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

# Accessing the System

## KVM System Bay 1



Front



Rear

## Service Tray System Bays 2-8



Front



Red Ethernet  
Cable



IEC C13  
Plug

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

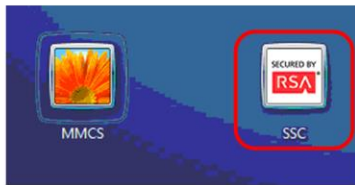
VMAX3 Hardware Fundamentals

73

The Keyboard, Video, Mouse (KVM) component is located in System Bay 1 of single engine and dual engine systems. The KVM provides service access to Simplified Symmwin for onsite service activities.

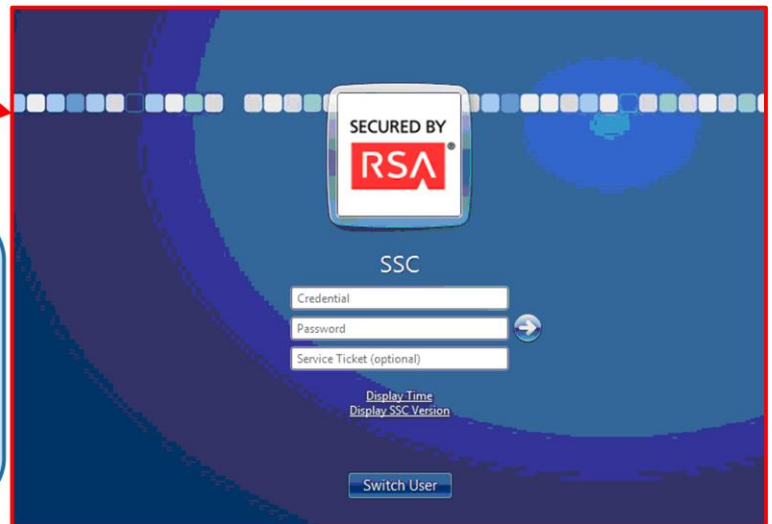
A Work Tray is located in Bays 2 through 8 in the VMAX<sup>3</sup> Family arrays. The Work Tray provides connectivity for a laptop for remote access to scripted procedures from any bay other than System Bay 1. This can be especially helpful when servicing bays in a large data center or on a dispersed system. A red Ethernet cable and standard 220 volt IEC C13 power cable are attached to the Work Tray to attach a service laptop to the system. Service personnel must supply an adapter plug or cable for connectivity to the power connector. An IEC C14 connector is attached to connect the cable to the tray. A NEMA 5-15R connector is needed to connect to the laptop. This connector is standard on all North American laptops. Other locations may require an IEC C14 to NEMA 5-15R adapter for connectivity. These adapters must be purchased separately for the CE Tool Kit. Do not leave these adapters onsite, as they are part of the CE Tool Kit.

## Accessing the System - Logon



### Log in using your Secure Credentials

- CE will auto login to Simplified Symmwin
- Eng will auto login to SymmWin



EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved.

VMAX3 Hardware Fundamentals

74

When accessing the system, at the Window Login screen, choose SSC and enter you Secure Credential, Password and Service Ticket. Upon login the appropriate GUI will be opened. For example; when providing CE credentials, Simplified SymmWin will automatically be launched and when providing Engineering credentials, Symmwin will automatically be launched.

# Simplified Symmwin

The screenshot displays the Simplified Symmwin interface. On the left is a login window titled 'Simplified Symmwin' with fields for 'User' and 'Password', and buttons for 'Login', 'Cancel', and 'Help'. The main window shows the 'Site Note' and 'Tasks' views. The 'Site Note' view has a text area and an 'Add Note' button. The 'Tasks' view shows a table of tasks with columns: SR, Part Na, Part Type, MM, Ser, Sta, Rep, Rep, Exe, Res. A callout 'Task Detail' points to the 'Part Type' column. Below the table are buttons: Run, Resume, Abort, Add, and Remove. At the bottom, there are tabs for 'Site Info', 'Lock Management', 'Replacement Logs', and 'System Management'. A callout 'Buttons for additional View Blocks' points to these tabs. Below the tabs, 'User and Role' is displayed as 'User: 3 Role: ENG / Engineering Support'. A callout 'Task Actions' points to the 'Add' button. The bottom of the screen features the EMC logo, copyright information, and the page number 75.

Serial Number: 196700571 Primary MMCS-1 Franklin 55 Lab Code: 5977.200.156.4 id=0

Site Note 02:19:04 PM Add Note

Tasks 02:19:04 PM

| SR | Part Na    | Part Type | MM | Ser | Sta  | Rep | Rep | Exe | Res  |
|----|------------|-----------|----|-----|------|-----|-----|-----|------|
|    | ProcEntry( |           |    |     | Fail | 06/ |     | 099 | true |

Run Resume Abort Add Remove

Site Info Lock Management Replacement Logs System Management

User: 3 Role: ENG / Engineering Support Server:

EMC<sup>2</sup>

Copyright © 2014 EMC Corporation. All Rights Reserved. VMAX3 Hardware Fundamentals 75

Upon login to Simplified Symmwin, *Site Note* and *Tasks* “View Blocks” are displayed. Across the bottom left, you will see additional buttons for View Blocks that can be opened - *System Management*, *Site Info*, *Lock Management* and *Replacement Logs*. On the lower right of the screen, User and Role are displayed. Depending on the user’s access level, options will differ. In this example, we’re logged on as ENG/Engineering Support.

## Module 4: Summary

Key points covered in this module:

- Access a VMAX<sup>3</sup> array for service

This module covered accessing the VMAX<sup>3</sup> with HYPERMAX OS 5977 arrays.

## Course Summary

Key points covered in this course:

- List the VMAX<sup>3</sup> models
- Explain the VMAX<sup>3</sup> hardware architecture
- Describe the VMAX<sup>3</sup> System Bay Components
- Accessing the system

**This concludes the training; proceed to the course assessment.**

After launching the assessment, you must complete it before returning to the course.

The course will automatically move to your Transcript within 48 hours after passing the assessment.



This course covered the list of the VMAX<sup>3</sup> models, explained the VMAX<sup>3</sup> hardware architecture, described the VMAX<sup>3</sup> System Bay Components, and how to access the system.

This concludes the training.